NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Junior C++ Systems Developer • Zerodha (Rainmatter)

+91 6305761091 • saratchandranallabati@gmail.com • Portfolio: sarat-2007.github.io • GitHub: github.com/sarat-2007 • LinkedIn

Education & Academic Pedigree

Amrita Vishwa Vidyapeetham **August 2024 – Expected May 2028**

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort) *Amritapuri, India*

- **Academic Distinction: AP POLYCET State Merit Rank: 497** (Top 0.5% statewide percentile across 100,000+ examinees).
- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Real-Time C++, ROS 2, Embedded Systems, Deep Learning, Classical Mechanics.

Peer-Reviewed Publications & Research

- **N. V. D. S. Sarat Chandra**, M. Nunna, Asha C. A., P. C. Sreekumar, **"Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity."** in *Proc. IEEE ICRM*, 2025. (First Author – Architected OpenCV 35+ FPS autonomy with 70% BOM cost reduction).
- **N. V. D. S. Sarat Chandra**, et al., **"Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation."** in *IEEE IES*, 2026. (Finalist – Formulated SE(3) equivariant diffusion with Hamiltonian energy conservation; 3.2x faster convergence).
- **N. V. D. S. Sarat Chandra**, et al., **"Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty."** Submitted to *IEEE Transactions on Industrial Informatics*, 2026. (120k samples, Optuna ensemble, 95% coverage).

Key Engineering Projects & Systems Architecture

Autonomous Mobile Robot (AMR) Edge Platform (IEEE ICRM 2025) **First Author & Lead Developer**

ARM Cortex-A, OpenCV, Probabilistic Hough Transforms, Asynchronous UDP Sockets

- 35+ FPS lane boundary tracking; <15ms perception-to-actuation telemetry latency.

Evolution Edge: Self-Evolving Heterogeneous Neural Bridge **Team Lead, AMD Developer Hackathon**

AMD Ryzen AI NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, INT8 PTQ, PEFT/LoRA

- Sub-28.4ms deterministic inference @ 8W TDP; offloaded tensor compute across integrated GPUs/NPUs.

Pasupatha Autonomous Companion Robot Control Bus **Head of Embedded Systems, Club Mirai**

ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed

- 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

Honors, Distinctions & National Merit

- **ISRO-IIRS YUVIKA Space Science Scholar:** Selected among top nationwide students for residential aerospace and satellite telemetry training.
- **NAEST National Physics Test Winner:** Qualified Prelims and Finals in experimental and conceptual physics (IAPT / Prof. H.C. Verma).
- **MTSO National Olympiad Level 2 Qualifier:** National mathematics and algorithmic reasoning distinction.

Technical Skills & Core Competencies

Programming Languages: C++ (C++17/C++20), Python (3.12), C (C99/C11), Bash, SQL, POSIX Sockets, Multi-Threading

Software Architecture & Design: Object-Oriented Design (OOP), Data Structures & Algorithms, REST APIs, Microservices, Asynchronous UDP/TCP Sockets

Robotics & AI Infrastructure: ROS 2 (Jazzy), CycloneDDS, OpenCV (35+ FPS), PyTorch, ONNX Runtime, Vitis AI EP, Zero-Copy Shared Memory

Tooling & CI/CD Systems: Linux (Ubuntu Server), Git/GitHub, Docker, CI/CD Pipelines, PyTest, GDB, Valgrind, CMake, Tectonic LaTeX